

Strengths and limitations of each approach

1. SARIMAX
 - a. Strengths
 - SARIMAX is very good in Seasonal Modeling and it captures trend and seasonal patterns effectively
 - It works well with Small Datasets and does not require large training data.
 - b. Limitations
 - Cannot model complex nonlinear relationships in data
 - Very Sensitive to Parameter Selection and can significantly reduce performance.
 - Cannot capture very long patterns.
2. XGBoost
 - a. Strengths
 - Learns complex feature interactions better than statistical models.
 - Very quick in Training and inference, even with large datasets
 - XGBoost model is robust against Overfitting and has built-in regularisation
 - b. Limitations
 - XGBoost requires Heavy Feature Engineering and lag features like rolling statistics to capture temporal behavior.
 - It has No Sequential Memory and Cannot naturally understand sequence order like recurrent networks.
3. LSTM
 - a. Strengths
 - Excellent Sequential Learning and captures temporal dependencies from sequential data.
 - Strong Long-Term Memory Handling
 - Learns long-range relationships effectively.
 - Learns patterns directly from raw sequences.
 - b. Limitations
 - High Computational Cost
 - Training is slower and often requires GPU acceleration.
 - Needs Large Training Data
 - Performance may degrade on smaller datasets.

Conclusions

- a) SARIMAX is best for seasonal forecasting and performs well when the data contains trends and seasonality, but does not perform well with nonlinear patterns.
- b) XGBoost provides the best balance between speed and accuracy, and with proper feature engineering, it achieves good performance and fast inference for practical deployment.
- c) LSTM is suitable for complex sequential forecasting. It effectively captures long-term temporal dependencies and nonlinear behaviour.

Possible Improvements to achieve better overall forecasting accuracy

1. Improve Feature Engineering
 - Add advanced lag features, rolling statistics, Fourier seasonal terms, and interaction features to improve SARIMAX and XGBoost performance.
2. Optimise Hyperparameters
 - Use techniques such as grid search to tune parameters
 - SARIMAX orders
 - XGBoost tree parameters
 - LSTM architecture and learning rates.
3. Develop Hybrid Models
 - Combine:
 - i. SARIMAX for seasonality
 - ii. XGBoost for nonlinear residual correction
 - iii. LSTM for deep temporal learning